
What does self-supervision add to engineered features for stellar light curves?

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Abstract

Self-supervised models promise one representation, pre-trained once on unlabelled survey data and read out cheaply for many downstream tasks. In stellar time-domain astronomy that promise competes with has to improve on a strong, cheap incumbent: a few dozen hand-engineered light-curve statistics. We pre-train a Conv1D variational encoder with a recurrent latent-dynamics model on 195,883 unlabelled TESS two-minute light curves and freeze it. We then evaluate, freeze it, and ask whether the features it has learned for each star learns add anything to 25 engineered features under one linear readout, on ten labelled tasks spanning variability classification, asteroseismology and rotation. Each task is scored against three baselines: the engineered features alone, an architecture-matched supervised Conv1D trained end-to-end, and an MLP on raw flux. Concatenating the learned and engineered features improves a frozen linear readout beyond 2 SE on 7 of 10 tasks, while ; an untrained encoder does not. The same frozen pipeline Against two networks trained end-to-end on the same labelled stars, splits and inputs, the frozen readout is ahead of the a supervised Conv1D with the encoder's architecture on 6 of 10 tasks and indistinguishable from it on the other 4, and ahead of the raw-flux MLP on an MLP on raw flux on 9 of 10. Self-supervision adds to engineered features rather than replacing them. Combined under Under a plain logistic regression, the combination matches or exceeds the two match end-to-end deep supervised networks trained we train on the same labelled stars, 669 to 16,002 labelled stars per task.

1 Introduction

Wide-field photometric surveys now produce light curves far faster than they can be labelled. TESS [Ricker et al., 2015] has observed hundreds of thousands of bright stars at two-minute cadence, and LSST [Ivezić et al., 2019] will enlarge the gap by orders of magnitude. Self-supervised learning is the natural response: pre-train one encoder on the unlabelled archive, then attach a cheap readout for each scientific question. This shape has recently been pursued for astronomical light curves by Donoso-Oliva et al. [2025], Zuo et al. [2025] and Rui [2026], and for TESS by Poznanski [2026], who validates with embedding diagnostics rather than downstream tasks.

What makes this domain unusual is the strength of the incumbent it must improve on. Decades of variable-star research have distilled light curves into compact engineered summaries – periodogram peak structure, band-limited power, spectral entropy, robust scatter statistics – and pipelines built on them remain highly competitive [Audenaert et al., 2021]. The benchmark of Li et al. [2025] makes the point sharply: across seven expert-labelled variable-star classes, no time-series foundation model they tested strictly surpassed the long-established domain feature baseline on classification. A learned representation for this data therefore has to answer a specific question, not a generic one: does it carry information the engineered features do not already have?

We take that question literally and measure it. We pre-train a latent-dynamics encoder on unlabelled TESS photometry, freeze it, and ask what the features it learns add to a 25-feature engineered stack under an identical linear readout – and whether the resulting pipeline, which trains no deep network at readout time, *is competitive with deep matches or exceeds two* networks trained end-to-end on the same labels. Our contributions are:

- **The learned features are complementary to the engineered ones, not a replacement for them.** Concatenating the two improves a frozen linear readout beyond 2 SE on 7 of 10 tasks, with the gains concentrated where the engineered basis is information-starved; an untrained-encoder control shows the effect is not generic column addition.
- **The combined pipeline, read out by logistic regression, *is competitive with matches or exceeds the two supervised networks we train end-to-end deep learning on these the same labels.*** Against *an architecture-matched* a supervised Conv1D *with the encoder’s architecture* it is ahead on 6 of 10 tasks and indistinguishable on the other 4; against an MLP on raw flux it is ahead on 9 of 10.

2 Data and method

Data. We use TESS two-minute SPOC PDCSAP photometry for 195,883 stars brighter than $T = 10$. Each light curve is split into contiguous segments at runs of missing cadences and at time gaps exceeding five times the median cadence; segments are median-normalised by their median absolute deviation and sliced into non-overlapping windows of 256 cadences (≈ 8.5 h). Missing values are never interpolated or padded – any window containing one is discarded, and segments are never stitched across sector or gap boundaries. Pre-training consumes sequences of 16 consecutive windows (≈ 5.69 d) and uses no labels of any kind.

Encoder. A 1D convolutional variational encoder [Kingma and Welling, 2014] maps each window to a 128-dimensional Gaussian posterior, and a GRU [Cho et al., 2014] predicts the next window’s latent from its predecessors, run both forward and backward over the sequence. The training objective combines window reconstruction, the KL term, the latent-dynamics prediction error, and a Hann-tapered spectral auxiliary term that penalises reconstruction error in the log periodogram. The encoder is trained once with six random seeds and then frozen; no gradient from any label ever reaches it.

Representations and readout. A star’s learned features are μ , the mean over its first segment (16–20 windows) of the per-window posterior means. The engineered baseline is a fixed 25-feature vector modelled on the T’DA variability pipeline [Audenaert et al., 2021]: the three strongest periodogram peaks with their normalised powers and ratios, a harmonic-power ratio, power in eight log-spaced frequency bands, spectral entropy and centroid, skew, kurtosis, point-to-point scatter ratio, and three robust dispersion statistics. Readouts are deterministic and frozen, standardised on the training split alone: class-balanced L_2 logistic regression at a fixed $C = 1$ for the detection tasks, ridge for the two regressions with its penalty set by leave-one-out cross-validation. *Cross-validating (cross-validating C as well changes the reported gains by at most moves no reported gain by more than 0.011, and the two tasks that move by more than 0.008 both move in the fusion arm’s favour).* We compare four arms, each the same readout on a different feature set: **features**, μ , **features** $\oplus \mu$, and, as a control, the same fusion built on an *untrained* encoder. The two supervised baselines of the next paragraph are the fifth and sixth arms.

Supervised baselines. Two end-to-end baselines see exactly the same input windows, star sets and splits. **C1** reuses the encoder’s convolutional trunk verbatim, adds a bottleneck and a scalar head, and is trained per task with a star-level bag loss (the mean of the star’s window logits, taken in-graph) and class weighting. **C2** replaces the convolutional trunk with a dense one and changes nothing else, so **C1** versus **C2** isolates the convolutional inductive bias. Both use one pre-registered regularisation setting for all tasks, select on a validation split no probe has ever touched, and are trained with *three six* seeds.

Tasks and metrics. We score ten tasks with one probe per physical quantity: pulsating variables [Gao et al., 2025], eclipsing binaries [Prša et al., 2022], rotation and rotation period [Boyle et al., 2026], transits [Guerrero et al., 2021], flare activity [Seli et al., 2025, Vida et al., 2021], oscillating red giants and their $\nu_{\max}$ [Hon et al., 2021], solar-like oscillators [Hatt et al., 2023], and RGB versus

Table 1: Ten-task scorecard. Linear-readout arms use a frozen logistic or ridge readout on the mean-pooled representation; the two supervised arms are trained end-to-end per task. Bold marks a leader separated from the runner-up by the combined 2 SE band; italic marks a leader that is only the point-estimate armax. n is the test-star count and prev. the positive-class prevalence.

task	metric	n (prev.)	frozen linear readout			end-to-end supervised	
			features	μ	features $\oplus \mu$	Conv1D	MLP
pulsating	PR-AUC	2021 (0.107)	0.789	0.806	0.846	0.816	0.748
eclipsing binary	PR-AUC	2021 (0.097)	0.742	0.771	<i>0.775</i>	0.774	0.719
rotation	PR-AUC	2021 (0.089)	0.540	0.559	<i>0.555</i>	<i>0.565</i>	0.526
transit	PR-AUC	2021 (0.060)	0.190	0.144	0.218	0.134	0.109
oscillating giant	PR-AUC	3429 (0.383)	0.920	0.854	0.918	0.875	0.883
solar-like osc.	PR-AUC	3429 (0.122)	0.320	0.336	0.390	<i>0.396</i>	0.275
RGB vs. HeB	ROC-AUC	161 (0.702)	0.758	0.660	0.708	0.530	0.762
$\nu_{\max}$	R^2	1313	0.831	0.802	0.867	0.832	0.834
rotation period	R^2	150	0.703	0.677	0.717	0.597	0.698
flare activity	PR-AUC	3429 (0.089)	0.474	0.453	0.524	<i>0.533</i>	0.444

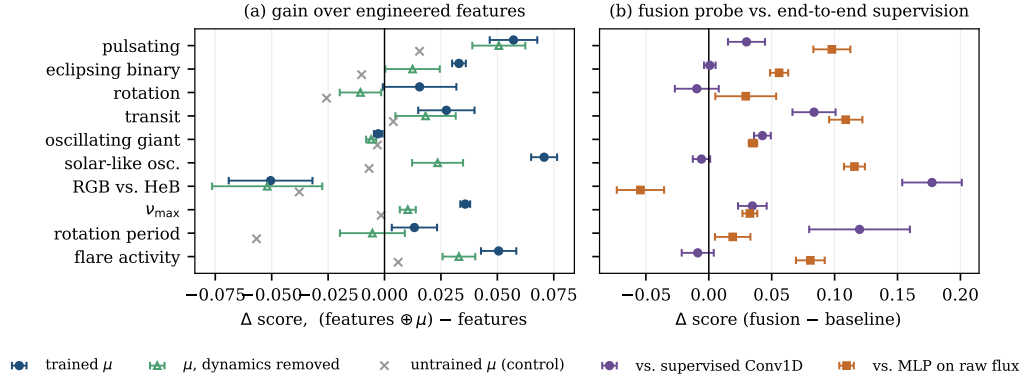


Figure 1: The two claims. **(a)** What the learned features add to the engineered features under an identical frozen linear readout, for the shipped encoder, the same recipe with the latent-dynamics term removed, and an untrained-encoder control. **(b)** The frozen fusion probe against the two end-to-end supervised baselines. Bars are 2 SE; the untrained control is a single encoder and carries none.

helium-burning discrimination [Sreenivas et al., 2026]. Here $\nu_{\max}$ is the frequency of maximum oscillation power. Detection tasks are scored by the area under the precision–recall curve (PR-AUC), whose floor is the positive-class prevalence rather than 0.5; prevalence is therefore quoted with every detection score. The two regressions are scored by R^2 .

Uncertainties. All error bars are $2\times$ the standard error of the mean over seeds. For the μ arms the seeds are independent pre-training runs (six); for C1 and C2 they are initialisation and shuffling seeds (six, matching the pre-training count). Because a pre-training seed and an initialisation seed are different objects, differences between a μ arm and a supervised arm are treated as unpaired and their bands added in quadrature even at equal counts. The engineered arm is deterministic and carries no seed spread. A difference counts as resolved only when it exceeds the combined 2 SE band. These bands measure re-training spread only; a paired bootstrap over the test stars (1,000 resamples, every arm rescored on the same stars, seeds averaged; Appendix C) excludes zero on 5 of the 7 fusion gains, 5 of the 6 leads over C1 and 7 of the 9 over C2, the rest stay ahead in at least 65% of resamples, none changes sign, and the rows that loosen are the two 150–161-star tasks.

3 Results

The learned features add to the engineered features. Table 1 and Figure 1a give the comparison. Concatenating μ with the 25 features improves the frozen readout beyond 2 SE on 7 of 10 tasks, led by solar-like oscillators +0.071, pulsating +0.057 and flare activity +0.051, and reaching +0.013 (R^2) on rotation period. Rotation is positive but just short of the bar (+0.016 against 0.016). The two negative tasks are the ones where the engineered basis is already saturated or the probe is small: oscillating giants (−0.003, on a baseline of 0.920) and RGB versus helium-burning (−0.051, 161 test stars). On both, an untrained-encoder control loses by an amount we cannot resolve from the trained fusion’s – the signature of extra collinear columns diluting a linear fit, not of μ carrying misleading content; Appendix A gives both rows in full. Across all ten tasks the untrained control is negative on seven and never exceeds +0.016, so the gains are specific to what pre-training learned. μ on its own does not replace the features: it is ahead on only 4 of 10 tasks alone, which is the same conclusion Li et al. [2025] reach for foundation-model embeddings, and is why we report fusion rather than substitution [Makinen et al., 2024].

The frozen pipeline is competitive with end-to-end deep learning matches or exceeds the two supervised networks. Against C1 – the same convolutional trunk, the same labels, the same input windows, trained end-to-end – the frozen fusion probe is ahead on 6 of 10 tasks and indistinguishable from it on the other 4 (Figure 1b). Against C2, an MLP on raw flux, it is ahead on 9 of 10 and behind on one (RGB versus helium-burning, 161 stars). C1 is not weak: it clears the engineered arm on 5 of 10 and the μ -only arm on 4, and on eight seven tasks the frozen probe recovers 0.98–1.09 of its margin above the metric floor. The correct reading of this comparison is *not* that self-supervision matches supervision in general. It matches **This is a statement about** what this architecture reaches on the labels that actually exist, which range from 669 to 16,002 labelled stars per task, **not about supervision in general**.

The latent-dynamics term is what buys the complementarity. Replacing the forward–backward dynamics objective with a dynamics-free variant, all else fixed and the same six seeds, reduces the fusion gain on all 10 tasks, and by more than 2 SE on 6 when the two arms are differenced seed by seed (Figure 1a, open triangles; largest losses solar-like oscillators −0.047, rotation −0.026, $\nu_{\max}$ −0.025). The recurrent prediction term, not the autoencoder alone, supplies the information the engineered features lack.

4 Discussion

Model selection must be gated on a held-out probe. Model selection, not optimisation, is where this pipeline is fragile. The reconstruction objective develops a localised artifact at the window boundary, and none of eleven interventions across four families of fix satisfied both pre-registered acceptance criteria, so the tapered variant shipped on the stated rule, not because it won. Underlying this is a measurement we think generalises one measurement (Appendix B). Across ten pre-training recipes of one sweep, validation reconstruction *anti*-predicts downstream probe score: the Spearman ρ between negated validation reconstruction and PR-AUC runs from −0.58 to −0.84 over four tasks, the loss-optimal recipe is the dynamics-free one on all four, and trusting the loss costs 0.021–0.049 PR-AUC. The effect lives across recipes, not across seeds of one recipe, where no label-free metric we tried ranks anything. Self-supervised representation work in this domain therefore cannot be tuned on its own objective and must be gated on a held-out probe.

Limitations and next steps. The four linear-readout arms (features, μ , features $\oplus \mu$, and the untrained control) share one linear model. It has few parameters, so it cannot rescue weak features, and it is identical across the four, so the comparison is between feature sets. Our claim is made for these 25 engineered features and a linear readout, and ; we make no claim about other feature sets or more powerful readouts. Our corpus is the bright ($T < 10$) two-minute sample, and every arm, the two supervised baselines included, sees one 5.7-day segment per star. The fainter and , far more numerous stars , and longer and light curves spanning several TESS sectors , are the obvious next experiments; the latter would let rotation periods longer than 5.7 days be measured, which one segment cannot do. Our negatives are stars absent from the label catalogues, not stars verified to lack the signal, so a correct detection the catalogue missed counts against us; every detection score is therefore a lower bound. The flare label marks stars that have flared somewhere in their TESS data,

not a flare inside the 5.7-day segment we encode, so the task is recognising a flaring star, not finding a flare. When the readout is trained on one percent of the labels, the advantage narrows on 6 of 10 tasks, so how much pre-training helps when labels are scarce remains an open question.

Reproducibility and disclosure. §2 gives the corpus cut, geometry, features, objective and readouts; the remaining sizes are encoder has four convolutional stages of width 32/64/128/256, (kernel 5, each halving length), a 128-dimensional Gaussian latent, and a single-layer GRU of width 256 across the 16 windows. 256. Splits are by star, drawn once, and shared byte-for-byte by every arm, including both supervised baselines, which reproduce the probe’s star sets and test counts exactly the supervised baselines included. Code and score tables will be released on publication; generative AI assisted with analysis code and prose editing.

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A The two tasks where fusion does not help

task	features	fusion Δ	2 SE	untrained Δ	n_{test}
oscillating giants	0.920	−0.003	0.002	−0.003	3,429
RGB vs. He-burning	0.758	−0.051	0.019	−0.038	161

Oscillating giants: the engineered basis is already near-sufficient. The labels of Hon et al. [2021] come from a convolutional classifier reading a 128×128 image of each star’s power spectrum, which fires on a visible oscillation power excess at $6 \lesssim \nu_{\text{max}} \lesssim 250 \mu\text{Hz}$. Our engineered vector encodes close to that same quantity: eight log-spaced band powers spanning 0.3–360 c/d, plus the spectral centroid and entropy, locate the excess and describe its concentration. The features accordingly reach 0.920, and the trained and untrained fusion arms then lose the same 0.003 – 128 collinear columns diluting a linear fit, with no trained-versus-untrained difference left to attribute to μ .

RGB versus helium-burning: no arm resolves the discriminant. Sreenivas et al. [2026] assign the evolutionary state with a convolutional classifier reading the background-corrected power spectrum *folded at $\Delta\nu$* ; the physics underneath is the dipole mixed-mode period spacing $\Delta\Pi_1$ [Bedding et al., 2011]. Neither is accessible at our baseline. The encoder sees one 256-cadence window at a time, so the finest frequency structure available to μ is $\Delta f = 32.6 \mu\text{Hz}$, while $\Delta\nu$ for these stars is $\approx 1.5\text{--}18 \mu\text{Hz}$ – the comb itself is unresolved, let alone a fold of it. The engineered features, computed on the whole 5.69 d segment, reach $2.0 \mu\text{Hz}$, still coarser than the mixed-mode spacing $\Delta\Pi_1\nu^2 \approx 0.2\text{--}0.6 \mu\text{Hz}$ near $\nu_{\text{max}} = 50 \mu\text{Hz}$. No arm on this row measures the quantity the label encodes; the 0.758 the features reach rides on correlates of evolutionary state rather than on its discriminant. The row behaves accordingly: it is the only one where C2 (0.762) is ahead of the frozen probe while C1 reaches only 0.530 on 755 training stars, and it carries the smallest test set in the study.

237 B Validation reconstruction against downstream score

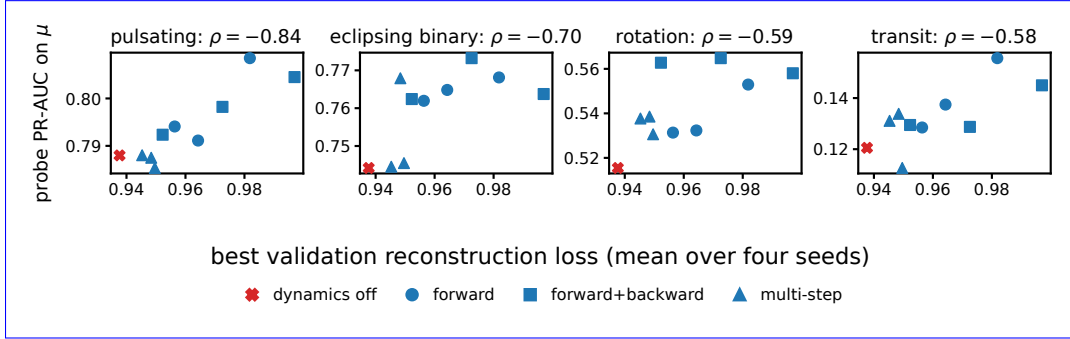


Figure 2: One point per pre-training recipe (mean over its four seeds), one panel per task. The ten recipes share corpus, architecture and loss terms and differ only in the latent-dynamics term (off, forward, forward+backward, multi-step, the last three at three weights). x : best post-warm-up validation reconstruction loss; y : PR-AUC of the frozen linear probe on μ ; ρ : Spearman's between negated loss and score.

Figure 2 is the measurement behind the model-selection paragraph of §4. The loss-optimal recipe is the dynamics-free one (red cross) on all four tasks, and it is the worst or next-to-worst probe on each, so selecting on the loss costs 0.021–0.049 PR-AUC against the probe-optimal recipe. The trend is not carried by that one point: with it removed, ρ still runs from -0.43 to -0.87 . Two limits on what the figure shows. The recipes are an earlier sweep than the shipped encoder (they carry an earlier spectral auxiliary term), so the figure establishes that the anti-correlation exists in this family, not its size for the final recipe. And it is a statement across recipes only: within one recipe, across its seeds, no label-free curve metric we tried (reconstruction, reconstruction plus auxiliary, active latent units) ranks the seeds, every such correlation sitting within ± 0.3 of zero.

248 C Paired bootstrap over test stars

The 2 SE bands in the body come from re-training and say nothing about whether the particular test stars favoured one arm. For every printed contrast we therefore resample each task's test set with replacement 1,000 times, rescore every arm on the identical resampled list from its saved per-star predictions (no model is retrained), average each arm over its seeds, and take the difference. Table 2 gives the 95% percentile interval of that difference and the share of resamples in which the fusion arm is ahead; intervals that exclude zero are bold. The check tightens no verdict and reverses none: every row the body calls ahead has the fusion arm ahead in at least 65% of resamples. It does loosen the smallest tasks – rotation period (150 test stars) and RGB versus helium-burning (161) account for four of the ten rows where a resolved 2 SE verdict becomes an interval that touches zero.

task	fusion – features		fusion – C1		fusion – C2		fusion – dynamics-off fusion	
	95% CI	> 0	95% CI	> 0	95% CI	> 0	95% CI	> 0
pulsating	[+0.012, +0.105]	0.99	[−0.008, +0.073]	0.94	[+0.045, +0.150]	1.00	[−0.023, +0.036]	0.69
eclipsing binary	[+0.005, +0.062]	0.99	[−0.024, +0.025]	0.52	[+0.022, +0.094]	1.00	[+0.004, +0.038]	0.99
rotation	[−0.026, +0.057]	0.77	[−0.056, +0.040]	0.38	[−0.023, +0.079]	0.87	[−0.005, +0.052]	0.94
transit	[−0.025, +0.065]	0.85	[+0.041, +0.124]	1.00	[+0.046, +0.174]	1.00	[−0.016, +0.035]	0.78
oscillating giants	[−0.010, +0.005]	0.21	[+0.028, +0.057]	1.00	[+0.021, +0.051]	1.00	[−0.002, +0.008]	0.87
solar-like osc.	[+0.044, +0.097]	1.00	[−0.037, +0.022]	0.34	[+0.079, +0.151]	1.00	[+0.025, +0.067]	1.00
RGB vs. HeB	[−0.087, −0.012]	0.00	[+0.099, +0.247]	1.00	[−0.126, +0.020]	0.08	[−0.037, +0.039]	0.52
$\nu_{\max}$	[+0.028, +0.043]	1.00	[+0.024, +0.045]	1.00	[+0.018, +0.047]	1.00	[+0.020, +0.031]	1.00
rotation period	[−0.038, +0.067]	0.65	[+0.054, +0.186]	1.00	[−0.053, +0.100]	0.69	[−0.021, +0.060]	0.83
flare	[+0.019, +0.078]	1.00	[−0.047, +0.029]	0.32	[+0.027, +0.128]	1.00	[−0.000, +0.035]	0.97

Table 2: Paired star-bootstrap of the four contrasts the body counts. Each entry is the 95% interval of (left arm – right arm) over 1,000 resamples of the test stars, seeds averaged, and the share of resamples with the difference above zero. Bold: interval excludes zero.